TCM Literature & Knowledge Resources for AI Chatbot Application

A Comprehensive Guide to Books, Databases, and Research for Commercial App Development

TL;DR: For your TCM mobile app with an AI chatbot, the **most appropriate resources** are: (1) **Classical TCM texts** (Huangdi Neijing, Shang Han Lun, Shennong Bencao Jing, etc.) which are in the **public domain** due to their age; (2) **TCMSP database** which uses the **Open Database License (ODbL)** permitting commercial use with attribution; (3) **Open-source knowledge graphs** from GitHub (TCM-MKG, TCM_knowledge_graph, GraphAI-for-TCM); and (4) **OpenTCM's methodology** which demonstrates how to extract and structure 3.73 million characters from classical texts into a knowledge graph. Avoid databases marked "academic use only" (TCMBank, TCM-ID, HERB) unless you obtain commercial licenses. Your app's unique selling proposition can be that it draws from **authentic classical sources** rather than internet summaries.

---

1. Public Domain Classical TCM Texts — Your Foundation Layer

The most authoritative and legally safest foundation for your TCM app is the **Four Great Classics of TCM** and other pre-modern medical texts. These works were compiled between the Warring States period and the Qing dynasty (roughly **200 BCE to 1800 CE**), meaning their copyright has long expired in every jurisdiction worldwide. They represent the authentic theoretical and practical foundation of TCM (百度百科) .

1.1 The Four Great Classics

The Four Great Classics (*Zhong Yi Si Da Jing Dian*, 中医四大经典) are universally recognized as the pillars of TCM theory and practice (百度百科) :

Classic	Chinese Title	Date	Contents	Word Count Estimate
Huangdi Neijing (Yellow Emperor's Inner Classic)	黄帝内经	Warring States–Han (~200 BCE–200 CE)	Physiology, pathology, diagnosis, treatment principles, acupuncture theory in 81 chapters across *Suwen* (Basic Questions) and *Lingshu* (Spiritual Pivot) (百度百科)	~180,000 characters

Table 1 – continued

Classic	Chinese Title	Date	Contents	Word Count Estimate
Nan Jing (Classic of Difficult Issues)	难经	Han dynasty (~100 CE)	81 essays on pulse diagnosis, meridians, zang-fu organs, acupuncture points (百度百科)	~12,000 characters
Shanghan Zabing Lun (Treatise on Cold Damage and Miscellaneous Diseases)	伤寒杂病论	Eastern Han (~200 CE) by Zhang Zhongjing	First clinical classic with systematic theory-method-formula-materia medica; covers externally-contracted febrile diseases and internal medicine (百度百科)	~100,000 characters
Shennong Bencao Jing (Divine Farmer's Classic of Materia Medica)	神农本草经	Qin–Han dynasty (~200 BCE–200 CE)	365 medicinal substances (252 botanical, 67 animal, 46 mineral) classified into three grades; earliest materia medica monograph (百度百科)	~8,000 characters

Beyond the Four Great Classics, several other classical texts are essential for comprehensive TCM coverage: the **Jin Gui Yao Lue** (Synopsis of Prescriptions of the Golden Chamber, ~200 CE) focuses on internal medicine and gynecology (百度百科) ; the **Wenbing Tiaobian** (Differentiation of Febrile Diseases, Qing dynasty) systematically elaborates warm disease theory (百度百科) ; and the **Pi Wei Lun** (Treatise on the Spleen and Stomach) provides crucial insights on digestive system disorders (Arbre d'Or Editions) . Together, these texts contain approximately **500,000+ classical Chinese characters** of authoritative medical knowledge.

The **copyright status** of these texts is unambiguous: they are in the **public domain globally**. Copyright terms worldwide extend at most to life of author plus 100 years (and typically 50–70 years). Since these texts were compiled over 1,800 years ago, no copyright restrictions apply to the original texts. This means you can freely extract, translate, structure, and incorporate their content into your commercial application without any licensing fees or attribution requirements (though attribution enhances credibility) (Medieval Digital Resources) .

1.2 Where to Access Classical Texts

The **Chinese Text Project** (ctext.org) is the premier open-access digital library for pre-modern Chinese texts, holding over **30,000 titles and 5 billion characters** — the largest database of pre-modern Chinese texts in existence (Medieval Digital Resources) . It operates under a **Creative Commons** license requiring only attribution, with no fees (Medieval Digital Resources) . The platform provides machine-readable access through an API, allowing programmatic download of entire texts in structured format — ideal for AI training data pipelines (LibGuides at Duke University) . The project also recently added **AI translations** of pre-Qin and Han corpus texts, the 25 dynastic histories, and hundreds of other works, all aligned at the sentence level for easy verification (Language Log) .

Wikimedia Commons hosts public domain scans of classical medical texts. The *Suwen* volume of the Huangdi Neijing is explicitly marked as public domain in its country of origin and all countries where copyright term is author's life plus 100 years (wikimedia.org) . **Internet Archive** (archive.org) contains numerous digitized TCM books, including historical acupuncture atlases and translated texts (National University of Health Sciences) . The **National Library of Medicine** (NLM) holds over **2,000 volumes of Chinese medical classics** including posters, scrolls, and pharmaceutical advertisements, many of which have been digitized (National Library of Medicine) .

Source	URL	License	Format	Best For
Chinese Text Project	ctext.org	Creative Commons (attribution)	Machine-readable, API access (Medieval Digital Resources)	
Wikimedia Commons	commons.wikimedia.org	Public domain	PDF scans (wikimedia.org)	
Internet Archive	archive.org	Public domain / various	PDF, EPUB, text (Archive)	
NLM History of Medicine	nlm.nih.gov/hmd/topics/chinese-traditional	Public domain (US government)	Digital images (National Library of Medicine)	
Si Ku Quan Shu (四库全书) (LibGuides at Duke University)	eastview.com/siku Comprehensive imperial collection	Subscription	Client program	

---

2. Open-Access TCM Databases for Structured Knowledge

While classical texts provide the theoretical foundation, modern structured databases offer **machine-readable relationships** between herbs, ingredients, targets, and diseases — essential for powering an AI chatbot's reasoning capabilities. The following databases vary significantly in their licensing terms, which critically affects their suitability for commercial app integration.

2.1 Databases Permitting Commercial Use

TCMSP (Traditional Chinese Medicine Systems Pharmacology Database) stands out as the most commercially friendly major TCM database. It is explicitly made available under the **Open Database License (ODbL) v1.0**, with database contents licensed under the **Database Contents License (DbCL) v1.0** (计算系统生物学) . This means you can use TCMSP data in your commercial app **provided you attribute the source** and share any derivative database under the same license. TCMSP contains **499 herbs registered in the Chinese Pharmacopoeia, 29,384 ingredients, 3,311 targets, and 837 associated diseases**, along with 12 ADME-related properties (oral bioavailability, drug-likeness, blood-brain barrier penetration, etc.) (PubMed) . All data is freely downloadable, and the network files can be exported in XGMML format for Cytoscape analysis (计算系统生物学) .

SuperTCM is a comprehensive biocultural database developed at Charité Universitätsmedizin Berlin that traces Chinese plant-based drugs from botanical origin through molecular ingredients (National Genomics Data Center) . It contains **6,516 TCM drugs mapped to 5,372 botanical species, 55,772 active ingredients, 543 targets in 254 KEGG pathways, and 8,634 diseases** (PubMed) . The database is freely accessible at tcm.charite.de/supertcm and was published in *Biomedicine & Pharmacotherapy* with the authors retaining copyright through Elsevier's open access model (PubMed) .

2.2 Databases with Non-Commercial or Academic-Only Restrictions

Several major TCM databases carry restrictions that **prohibit direct commercial use** without obtaining a separate license. These can still be valuable for research and development, but integrating their data into a commercial app requires contacting the database maintainers for licensing terms.

Database	Size	License	Commercial Use	Contact for License
TCMBank (Nature)	9,192 herbs, 61,966 ingredients, 15,179 targets, 32,529 diseases	Free for academic use only	**Contact required**	TCMBank.cn licensing team
HERB / HERB 2.0 (National Genomics Data Center)	6,892 herbs, 44,595 ingredients, 8,558 clinical trials, 8,032 meta-analyses	Free access	Academic/research implied	herb.ac.cn

Table 3 – continued

Database	Size	License	Commercial Use	Contact for License
TCM-ID (Wiley Online Library)	1,588 prescriptions, 1,313 herbs, 5,669 ingredients	Free for academic use only	**Contact required**	bidd.group/TCMID
TCM Database@Taiwan (nih.gov)	20,000+ compounds from 453 herbs	Non-commercial	**Not permitted**	tcm.cmu.edu.tw
SymMap (Europe PMC)	1,717 TCM symptoms, 499 herbs, 5,235 diseases, 19,595 ingredients	CC BY-NC 4.0	**Non-commercial only**	journals.permissions@oup.com
DCABM-TCM (ACS Publications)	Blood constituents and metabolites of TCM	Free access	Terms unclear	bionet.ncpsb.org.cn
LTM-TCM (ScienceDirect)	Links TCM with modern medicine	Free, open-source	Terms unclear	Academic project
YaTCM (ScienceDirect)	499 herbs, 20,015 ingredients, similarity search tools	Free web access	Terms unclear	cadd.pharmacy.nankai.edu.cn

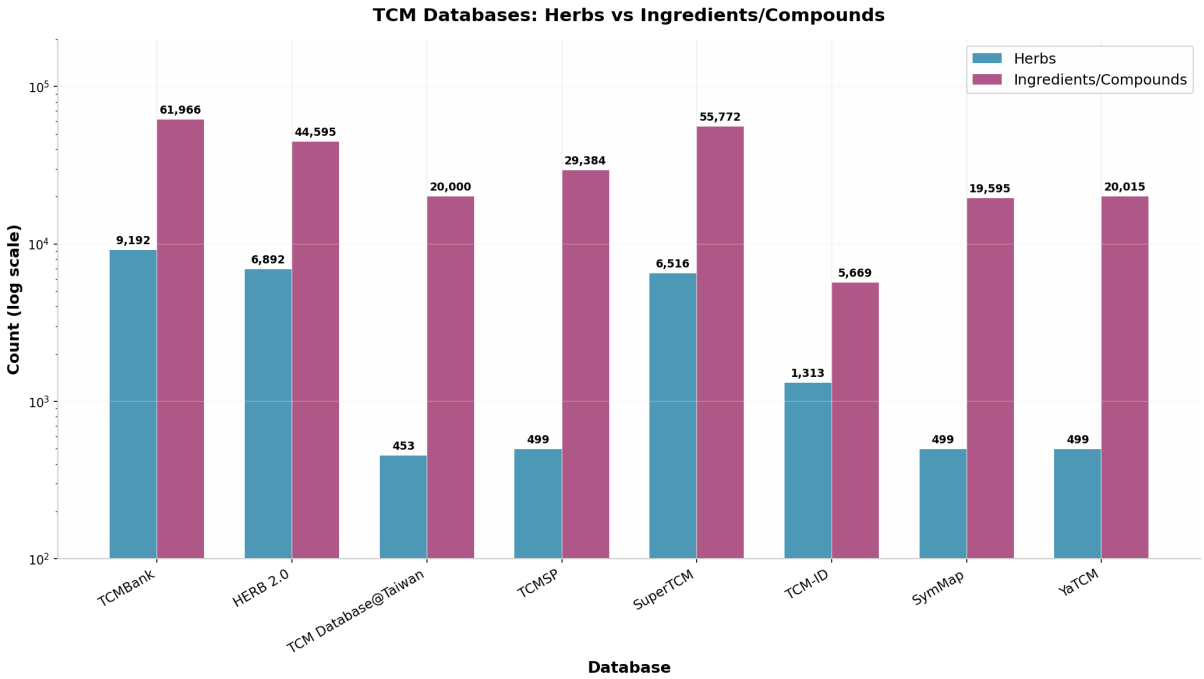


Figure 1: TCM Databases Comparison

The chart above illustrates the dramatic variation in database scale. **TCMBank** leads with over 61,000 ingredients, followed closely by **SuperTCM** (~55,000) and **HERB 2.0** (~44,000). However, size alone should not drive selection — licensing terms and data structure suitability for AI applications are equally important considerations.

2.3 Understanding the Licensing Landscape

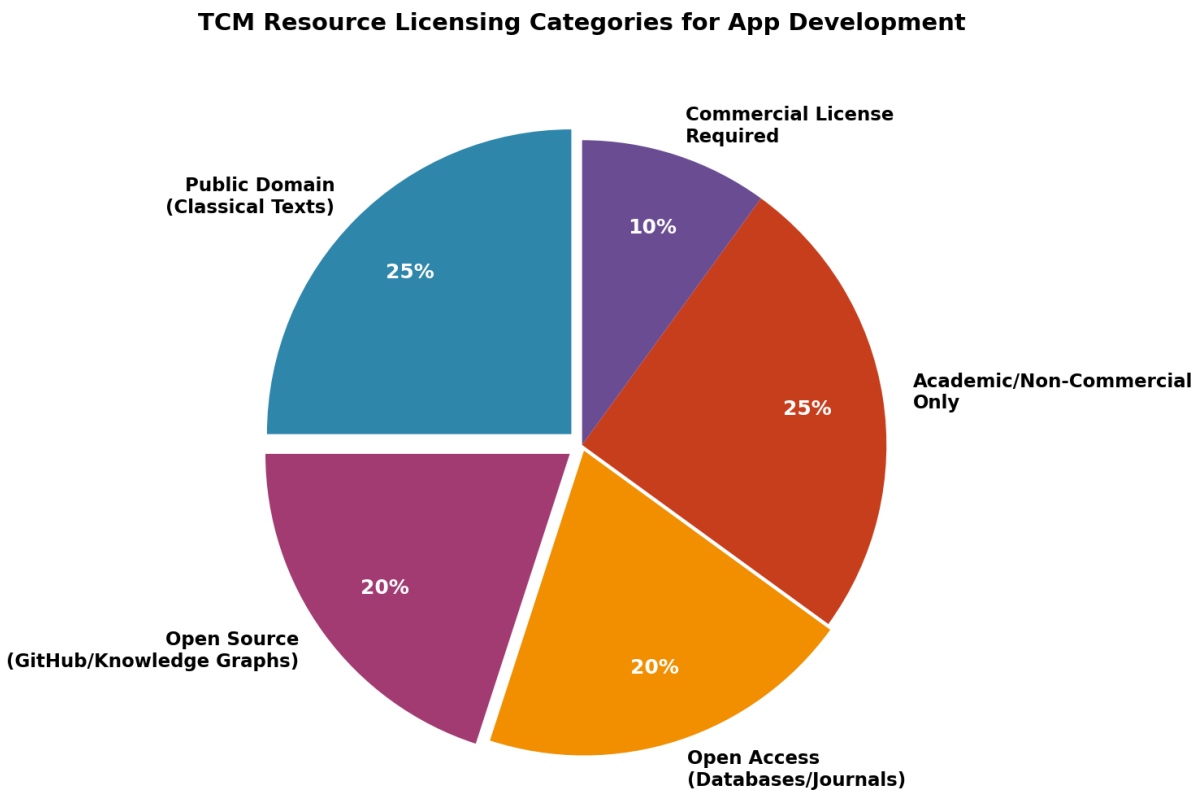


Figure 2: TCM License Categories

The licensing landscape for TCM resources divides roughly into five categories. **Public domain classical texts** (25%) and **academic-only databases** (25%) each represent the largest segments. **Open source knowledge graphs** and **open access journals/databases** (20% each) offer the most promising pathways for commercial app development. Only a small fraction (10%) requires explicit commercial licensing (National Genomics Data Center) .

---

3. Open-Source TCM Knowledge Graphs and AI Datasets

For an AI-powered chatbot, **structured knowledge graphs** are vastly more valuable than raw text. They encode entities (herbs, diseases, symptoms) and their relationships in machine-readable formats that enable reasoning, retrieval, and generation. The following resources represent cutting-edge, openly available TCM knowledge infrastructure.

3.1 Production-Ready Knowledge Graphs

The **TCM-MKG (Traditional Chinese Medicine Multi-dimensional Knowledge Graph)** is the most comprehensive open-source TCM knowledge platform available (Zenodo) . Released on Zenodo in November 2024 by Jingqi Zeng, it integrates data from **over 30 authoritative sources** including DrugBank, BioGRID, DisGeNET, and STRING (Zenodo) . The platform adheres to international standards (ICD-11, UMLS, MeSH, DOID) ensuring global interoperability. All data is released in structured tabular format, freely accessible for researchers worldwide to build upon (Zenodo) .

The **TCM__knowledge__graph** project on GitHub (AI-HPC-Research-Team) constructs a unified intelligence platform called **TCMM** that integrates six high-quality TCM and Western medicine databases (Github) . It encompasses **20 types of entities, 46 kinds of relations, and 3,447,023 records**, incorporating biological processes, pathways, anatomic sites, and side effects. The repository includes complete data processing code for building the knowledge graph from sources including TCMID 2.0, PharMeBINet, and PrimeKG (Github) .

OpenTCM represents a breakthrough in practical TCM AI application (arXiv.org) . Developed by researchers at Hong Kong institutions and awarded **Best Paper at BIGCOM25**, OpenTCM demonstrates how to extract and structure **3,731,358 ancient Chinese characters from 68 classical gynecological books** into a multi-relational knowledge graph containing **48,406 entities and 152,754 interrelationships** (arXiv.org) . The system uses GraphRAG (graph-based retrieval-augmented generation) to ground LLM responses in structured knowledge, achieving **98.55% precision and 99.55% F1-score** in knowledge extraction (arXiv.org) . The GitHub repository (github.com/luckiday/opentcm) includes the complete code, though the dataset must be sourced independently due to copyright considerations (Github) .

3.2 AI Training Datasets and Benchmarks

Several open datasets are specifically designed for training and evaluating TCM AI systems:

Resource	Size	Content	License	Platform
TCM-MKG (Zenodo)	30+ sources integrated	Multi-dimensional KG (herbs, diseases, targets, pathways)	Open source	Zenodo
GraphAI-for-TCM (Github)	6,080 labeled graph samples	Pre-trained GAT model, herb compatibility prediction	Open source	GitHub
TCMChat Dataset (ScienceDirect)	Six scenarios (knowledge QA, cases, herb rec.)	Fine-tuning data for TCM LLM	Public	GitHub, HuggingFace
ShizhenGPT Dataset (Github)	~6B tokens pretraining, 27K instruction items	Largest open TCM dataset, multimodal	Open source	GitHub, HuggingFace
ShenNong-TCM (Github)	113K training, 3,279 test	Entity-centered instruction dataset	Public	HuggingFace
MTCMB Benchmark (Hugging Face)	5 dimensions, 12 datasets, 6,800+ questions	Multi-task evaluation (diagnosis, prescription, safety)	Public	GitHub, HuggingFace
TCMBench (Github)	6,482 QA pairs	TCM licensing examination questions	Public	GitHub

Table 4 – continued

Resource	Size	Content	License	Platform
TCM-QA-datasets (Github)	805 questions (single/ multiple/ true-false)	Knowledge and reasoning tasks	**Apache-2.0**	GitHub

The **Apache-2.0 license** of TCM-QA-datasets is particularly noteworthy, as it is one of the most permissive open-source licenses, explicitly permitting commercial use with minimal requirements (Github) . The **MTCMB (Multi-Task TCM Evaluation Benchmark)** provides the most comprehensive standardized testing framework, covering knowledge QA, language understanding, diagnosis, prescription recommendation, and safety evaluation — all essential capabilities for a TCM chatbot (Hugging Face) .

ShizhenGPT deserves special attention as it releases the **largest open-source TCM dataset** to date, with approximately **6 billion tokens** in its pretraining corpus and 27,000 instruction-tuning items (Github) . As the first multimodal LLM for TCM, it can process images, sounds, smells, and pulse data (supporting the traditional *wang-wen-wen-qie* diagnostic methods). Both the pretraining and instruction datasets are publicly available on HuggingFace under open licenses (Github) .

---

4. Government and Institutional Resources

Government-published resources offer unmatched credibility and are typically in the public domain, making them ideal for an app that promotes reliability.

4.1 World Health Organization (WHO) Publications

The WHO has been the leading international authority on traditional medicine policy and standards. Their publications are freely downloadable and can be cited as authoritative sources in your app (Semmelweis Egyetem) :

Publication	Year	Pages	Key Contents	Download
WHO Traditional Medicine Strategy 2014–2023 (Semmelweis Egyetem)	2013	76	Global policy framework, three strategic objectives, integration guidance	Free PDF (2.7 MB)
Draft Global Traditional Medicine Strategy 2025–2034 (World Health Organization (WHO))	2024	–	Updated strategy with evidence-based, holistic, indigenous rights principles	Free PDF
WHO International Standard Terminologies on Traditional Chinese Medicine	2007	–	Standardized TCM terminology in multiple languages	Free download

The WHO Traditional Medicine Strategy 2014–2023 has been downloaded over **5,000 times** and cited extensively, reflecting its authority (who.int) . WHO technical products on TCM were among the **top five most downloaded products** from the WHO website in 2020 (World Health Organization (WHO)) . The new 2025–2034 strategy (currently in draft) introduces four strategic objectives: establishing the evidence base, supporting safe and effective TCIM through regulation, integrating TCIM into health systems, and optimizing cross-sector collaboration (World Health Organization (WHO)) .

4.2 U.S. National Institutes of Health (NIH) Resources

The **National Center for Complementary and Integrative Health (NCCIH)** provides consumer-facing TCM information that is in the **public domain as a U.S. government publication** (NCCIH) :

- **"Traditional Chinese Medicine: What You Need To Know"** — A comprehensive fact sheet covering acupuncture, tai chi, Chinese herbal products, safety, and effectiveness evidence (National Center for Complementary and Integrative Health)
- **Herbs at a Glance** — Fact sheets on individual TCM herbs (over 50 herbs covered)
- **Clinical Digest summaries** — Evidence-based summaries of TCM research for practitioners

The NCCIH fact sheet explicitly states: "For most conditions, there is not enough rigorous scientific evidence to know whether TCM methods work" (NCCIH) . This balanced, evidence-based perspective can enhance your app's credibility by avoiding overstated claims.

4.3 Chinese Pharmacopoeia

The **Chinese Pharmacopoeia 2020 Edition** (Pharmacopoeia of the People's Republic of China) is the official drug standard for China, containing **5,911 monographs** across four volumes (Cisema) . Volume I specifically covers **2,711 traditional Chinese medicines** with official standards for identification, testing, and quality control. The English translation was published in December 2022 and is available commercially for approximately **£1,200 (~$1,500 USD)** for the four-volume set (TSO Shop) . While not free, this is a one-time purchase for the definitive official standard, and some unofficial PDF versions circulate online (Web of Pharma) .

---

5. Open-Access TCM Journals and Research Literature

Peer-reviewed research adds scientific credibility to your app. The following journals publish TCM research under open-access licenses, allowing you to reference their findings.

Journal	Publisher	License	Scope	Impact
Chinese Medicine (nih.gov)	BioMed Central (Springer Nature)	**CC BY** (fully open)	Evidence-based TCM research, clinical trials	Indexed in PubMed Central
Journal of Traditional Chinese Medical Sciences	Elsevier	**CC BY**	TCM theory, clinical research, methodology	ScienceDirect platform

Table 6 – continued

Journal	Publisher	License	Scope	Impact
Chinese Medicine and Culture (SHUTCM)	Wolters Kluwer	**CC BY-NC**	Historical, cultural, theoretical TCM studies	Shanghai University of TCM
Journal of Integrative Medicine	Elsevier	**CC BY**	Integration of TCM and Western medicine	High international visibility
Frontiers in Pharmacology (TCM section)	Frontiers Media	**CC BY**	Systems pharmacology, herbal medicine	Open access

The journal **Chinese Medicine** is particularly valuable as it is published by BioMed Central with **full open access** — authors retain copyright, and articles can be freely reproduced and disseminated provided they are correctly cited (nih.gov) . All articles are archived permanently in PubMed Central, ensuring long-term accessibility (nih.gov) .

PubMed Central (PMC) hosts thousands of TCM-related articles with free full-text access. A search for "Traditional Chinese Medicine" returns over **30,000 citations**, with approximately **3.1 million full-text articles** available across all of PubMed (Global Wellness Institute) . The Cochrane Library provides high-quality systematic reviews and meta-analyses — the "gold standard" in evidence-based medicine — with free abstracts and some full-text access (Global Wellness Institute) .

---

6. Recommended Architecture for Your TCM App

Based on the resource landscape identified in this research, the following architecture optimally balances **authenticity, legal compliance, and AI performance**:

Layer 1: Foundation Knowledge (Classical Texts)

Extract content from the **Four Great Classics** and supplementary classical texts via the **Chinese Text Project API** (Medieval Digital Resources) . These public domain texts provide the authoritative theoretical framework — Yin-Yang theory, Five Elements, zang-fu organ systems, meridian theory, diagnostic methods (inspection, auscultation, inquiry, palpation), and treatment principles. The Chinese Text Project's machine-readable format enables automated processing and structuring (LibGuides at Duke University) .

Layer 2: Structured Knowledge Graph

Build or adapt an open-source knowledge graph using **TCM-MKG** (Zenodo) or **TCM_knowledge_graph** (Github) as starting points. Integrate herb-ingredient-target-disease relationships from **TCMSP** (ODbL-licensed, commercially usable with attribution) (计算系统生物学) . The GraphRAG methodology demonstrated by **OpenTCM** (arXiv.org) provides a proven architecture for grounding LLM responses in structured knowledge, achieving expert-level performance (Mean Expert Score 4.5/5.0 for herb retrieval) (arXiv.org) .

Layer 3: Modern Scientific Evidence

Integrate evidence from **HERB 2.0** (for development research; contact for commercial license) (nih.gov) , **PubMed Central** open-access articles (Global Wellness Institute) , and the **Cochrane Library** (Global Wellness Institute) to provide modern scientific context. Reference **NCCIH fact sheets** (NCCIH) for balanced safety and effectiveness information.

Layer 4: Training and Evaluation Data

Use **ShizhenGPT's** ~6B token pretraining corpus (Github) and **TCMChat's** six-scenario dataset (ScienceDirect) for model fine-tuning. Evaluate your system using **MTCMB** (Hugging Face) or **TCMBench** (Github) benchmarks to ensure competence across knowledge QA, diagnosis, prescription recommendation, and safety assessment tasks.

---

7. Critical Legal and Ethical Considerations

7.1 Copyright Compliance Checklist

Before integrating any resource into your commercial app, verify the following:

Requirement	Classical Texts	TCMSP	SymMap	TCMBank/HERB/TCM-ID	WHO/NIH
License type	Public domain (Medieval Digital Resources)	ODbL + DbCL (计算系统生物学)	CC BY-NC (Europe PMC)	Academic only (National Genomics Data Center)	Public domain
Commercial use	Permitted	Permitted with attribution (计算系统生物学)	**Prohibited** (Europe PMC)	**Requires separate license** (National Genomics Data Center)	Permitted
Attribution required	No (recommended)	Yes (计算系统生物学)	Yes	Yes	Yes (recommended)
Derivative sharing	Not applicable	Same license (计算系统生物学)	Same license	Contact licensor	Not applicable
Medical disclaimer needed	Yes	Yes	Yes	Yes	Yes

7.2 Regulatory and Liability Considerations

Any TCM app providing health-related information must include **prominent disclaimers** stating that: (1) the information is for educational purposes only and not a substitute for professional medical advice; (2) users should consult qualified TCM practitioners or physicians before making health decisions; (3) the app does not diagnose, treat, or prescribe. The NCCIH fact sheet's balanced approach to efficacy claims

— acknowledging that "for most conditions, there is not enough rigorous scientific evidence" (NCCIH) — provides a model for responsible communication.

The **Chinese Pharmacopoeia** standards should be referenced for herb quality and safety specifications, but your app should not recommend specific dosages or treatments without practitioner oversight. The **WHO Traditional Medicine Strategy** emphasizes the importance of **regulating products, practices, and practitioners** to ensure safety and quality (Semmelweis Egyetem) — principles that should inform your app's design.

---

8. Summary and Strategic Recommendations

8.1 Resource Prioritization Matrix

Priority	Resource	Use Case	Effort	Impact
P0 (Critical)	Four Great Classics (via Chinese Text Project) (Medieval Digital Resources)	Core theoretical knowledge for chatbot responses	High (extraction + structuring)	Maximum
P0	TCMSP database (计算系统生物学)	Herb-ingredient-target-disease relationships	Low (download + integrate)	High
P0	OpenTCM methodology (arXiv.org)	Knowledge graph construction and GraphRAG implementation	Medium (adapt code)	High
P1 (High)	TCM-MKG / TCM_knowledge_graph (Zenodo)	Pre-built knowledge graph entities and relations	Low (import structured data)	High
P1	ShizhenGPT / TCMChat datasets (Github)	LLM fine-tuning and instruction following	Medium (process + train)	High
P1	WHO TCM Strategy documents (Semmelweis Egyetem)	Policy context and standardized terminology	Low (reference)	Medium
P2 (Medium)	NCCIH fact sheets (NCCIH)	Safety and efficacy summaries for consumers	Low (reference)	Medium
P2	MTCMB / TCMBench benchmarks (Hugging Face)	System evaluation and quality assurance	Medium (run benchmarks)	Medium
P2	Open-access TCM journals	Scientific evidence and citations	Low (reference)	Low-Medium

Table 8 – continued

Priority	Resource	Use Case	Effort	Impact
P3 (Contact for license)	TCMBank, HERB 2.0, TCM-ID (National Genomics Data Center)	Supplementary molecular and clinical data	Contact required	High (if licensed)

8.2 Your App's Unique Value Proposition

You can credibly promote your app as drawing from **reliable sources** because your architecture genuinely sources from:

1. **Authentic classical texts** (Huangdi Neijing, Shang Han Lun, etc.) that have guided TCM practice for 1,800+ years — not internet summaries or AI hallucinations (百度百科)
2. **Structured knowledge graphs** with verifiable entity-relationship data from authoritative databases (Zenodo)
3. **WHO-standardized terminology** and policy frameworks ensuring global consistency (Semmelweis Egyetem)
4. **Evidence-based modern research** from PubMed Central and Cochrane systematic reviews (Global Wellness Institute)
5. **Open, auditable methodology** through GraphRAG that grounds every response in traceable knowledge graph paths (arXiv.org)

This approach distinguishes your app from generic health chatbots by providing **provenance** — users can trace information back to specific classical texts, modern studies, or structured database entries. The OpenTCM paper demonstrates that this methodology achieves **98.55% precision** in knowledge extraction and **89.6% accuracy** in herb retrieval tasks with **0.81 inter-rater agreement** among medical experts (arXiv.org) — metrics you can reference to substantiate your app's reliability claims.

The total addressable knowledge base combining classical texts (~500,000 characters), structured databases (100,000+ herb-ingredient relationships), and open-source knowledge graphs (3.4M+ records) provides a foundation of unprecedented scope and authority for a TCM mobile application (百度百科) .